RAG Complete Guide

What is RAG?

Retrieval-Augmented Generation combines retrieval and generation. Documents are indexed into a vector database and relevant chunks are retrieved before sending context to the LLM.

Pipeline

Loader → Chunker → Embeddings → Vector Store → Retriever → Prompt → LLM.

Chunking

Chunk size and overlap affect retrieval quality.

Embeddings

Embeddings map text into vectors so semantic similarity can be measured.

Vector Databases

Chroma, FAISS, Pinecone, Qdrant, Weaviate, and Milvus are common vector databases.

Retrievers

Similarity search retrieves the closest chunks. MMR balances relevance and diversity.